ETALE SCHEMES OVER A FIELD, AND THE FIBRES OF ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. The counted statement carries an author 'slogan' block and, just below it, a stray line 'Fibres of 'etale morphisms.' before the enumerate; both are retained verbatim because the card reproduces the author environment whole, and the slogan environment is a no-op defined in the chapter preamble.

Editorial note. The proof opens 'You can deduce this from Lemma ... via Lemma ... above. Here is a direct proof.' – a self-described alternative route immediately followed by the actual argument, so the span carries a complete proof and not a deferral; accepted with the wording quoted under the Lead's 12:4x rule.

Allowed standard prerequisites (editorial).

- Etale algebras over a field are finite products of finite separable field extensions. Algebra, Lemma \ref{algebra-lemma-etale-over-field}, cited at proof lines 8196-8197. Supplies the structure of each affine open of an etale scheme over the field.
- Points of a scheme whose residue field extension is algebraic are closed. Lemma \ref{lemma-algebraic-residue-field-extension-closed-point-fibre}, cited at proof lines 8202-8203. Shows the underlying space is discrete.
- Fibres of an etale morphism, and the description of an etale morphism by its fibres. Lemma \ref{lemma-unramified-over-field} and Lemma \ref{lemma-etale-smooth-unramified}, mentioned at proof lines 8193-8194. Alternative route the author mentions before giving the direct proof.

Source reference. Original lemma 02GL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3232–3232. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02G7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3218–3218. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3231–3231. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1097–1097. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01TE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2371–2371. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Fibres of étale morphisms.*

(1) *Let X be a scheme over a field k. The structure morphism $X \to \mathrm{Spec}(k)$ is étale if and only if X is a disjoint union of spectra of finite separable field extensions of k.*
(2) *If $f : X \to S$ is an étale morphism, then for every $s \in S$ the fibre X_s is a disjoint union of spectra of finite separable field extensions of $\kappa(s)$.*

Proof. You can deduce this from Lemma 02G7 via Lemma 02GK above. Here is a direct proof.

We will use Algebra, Lemma 00U3. Hence it is clear that if X is a disjoint union of spectra of finite separable field extensions of k then $X \to \mathrm{Spec}(k)$ is étale. Conversely, suppose that $X \to \mathrm{Spec}(k)$ is étale. Then for any affine open $U \subset X$ we see that U is a finite disjoint union of spectra of finite separable field extensions of k. Hence all points of X are closed points (see Lemma 01TE for example). Thus X is a discrete space and we win. □